Geophys. J. R. astr. Soc. (1970) **212,** 163–179.

Palaeomagnetism of Cretaceous Dikes from Jamaica

(Received 1970 August 20)

N. D. Watkins and F. W. Cambray

Summary

In order to contribute to resolution of hypotheses involving the tectonic history of the central Caribbean, a palaeomagnetic survey has been carried out on a series of Cretaceous dikes from the Central Inlier of Jamaica. The magnetic properties of the dikes, which are basalt, andesite, trachyandesite, and hornblende-andesite in composition, are variable but are generally dominated by low oxidation state titanomagnetites. This has caused some difficulty in defining a stable direction of magnetization. Nevertheless, two distinct palaeomagnetic groups are recognized: eleven dikes possess virtual geomagnetic poles (VGP) which when averaged are indistinguishable from the previously-defined North American Cretaceous pole position, and four other dikes recorded closely-grouped equatorial VGP. Both groups contain polarity differences. The favoured interpretation is that Jamaica has not rotated significantly about a vertical axis since the Cretaceous, and for a limited period the regional geomagnetic field behaviour resulted in low latitude virtual geomagnetic poles, similar to previously described geomagnetic activity detected in Tertiary lavas of Hawaii, Alaska, Oregon, and Iceland.

Introduction

The arcuate configuration of the islands of the Caribbean has inspired many imaginative hypotheses about the tectonic history of the region. Until recently they ranged essentially from the large and complex rotations and reconstructions of Carey (1958) shown in Fig. 1(a), to the large eastward bowing of an originally linear structure (Fig. 1(b)) as hypothesized, for example, by North (1965). Rod (1967), MacDonald (1968), and others, have assessed these published ideas.

The most recent summary of Caribbean tectonic hypotheses is by Gough & Heirtzler (1969), who emphasize the advance due to the Hess & Maxwell (1953) contribution. In this, Jamaica is envisaged as moving 950 km eastward south of the Cayman trough. Recent plate tectonic analyses (Molnar & Sykes 1969) support this proposal in that the Caribbean plate is inferred to be moving east with respect to North America and the west Atlantic. The Atlantic plate is thought to be underthrusting the Caribbean plate.

Because of the predictable high latitude of the Cretaceous or Tertiary geomagnetic pole positions with respect to North America, any longitudinal (east–west) movement is not readily detectable palaeomagnetically. Nevertheless, since the possibility of rotation about a vertical axis (particularly during Palaeozoic to early Tertiary times)

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

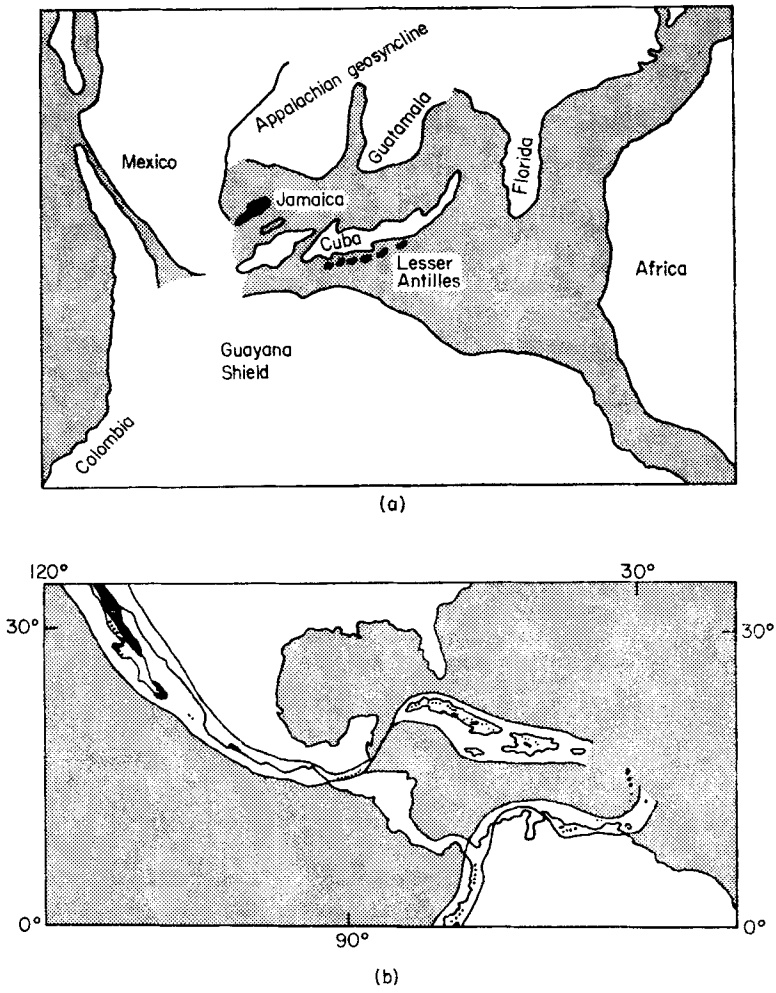


FIG. 1. Diagrams showing two tectonic hypotheses for the Caribbean: (a) is after Carey (1958). The reconstruction is envisaged for the Palaeozoic prior to the continental drifting which formed the Atlantic Ocean. (b) is after North (1965). The Antilles are envisaged as a bowed secondary feature, having originally been part of straight tectonic feature between North and South America. In both diagrams considerable rotations of the island of Jamaica are inferred.

is still not excluded, palaeomagnetic data are, as stressed by MacDonald (1968), of great potential value for resolution of Caribbean tectonic hypotheses. The lack of such studies is due in part to the general shortage of good unweathered volcanic outcrop on many of the islands.

It is the purpose of this paper to present palaeomagnetic data from Cretaceous dikes on the island of Jamaica. In addition to being relevant to the rotation hypotheses implicit in Fig. 1, it follows that the Cretaceous geomagnetic polarity history is examined in part. Helsley & Steiner (1969) have recently summarized and added to Cretaceous palaeomagnetic results: their analysis shows that the period is very dominantly normal polarity, agreeing with inferrences from oceanic magnetic anomalies over suspected Cretaceous oceanic crust (Heirtzler *et al.* 1968).

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

Geology of the area

The most comprehensive account of the geology of Jamaica is given by Zans *et al.* (1962).

The oldest proven rocks on the island are Cretaceous. They occur as a series of inliers (Fig. 2, centre) in the Tertiary limestones which rest with marked unconformity on the Cretaceous rocks. The largest of these inliers is the Central Inlier, although the most spectacular is the Blue Mountain Inlier to the east, which rises to 7375 feet above sea level. Speculation about the existence of rocks older than the Cretaceous has been made. The Blue Mountain metamorphic rocks have been been called Palaeozoic and thought of as a direct continuation of the Palaeozoic orogenic belt of Central America. Chubb (1962) presents an interesting discussion of what may lie below the Cretaceous rocks of Jamaica.

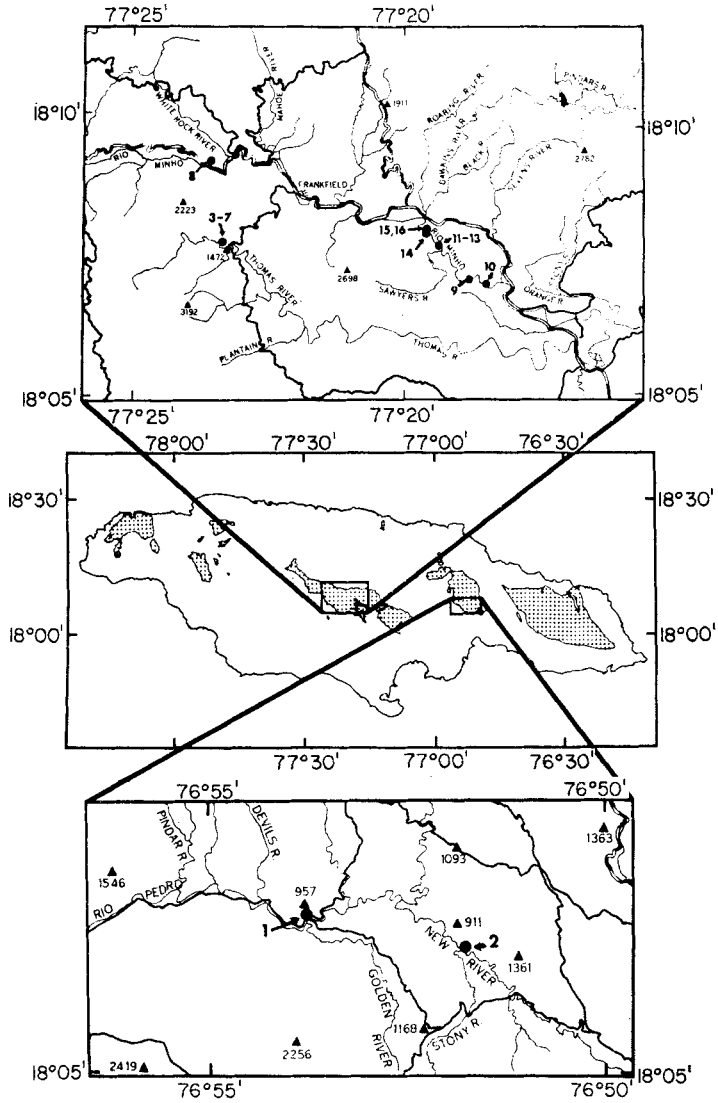


FIG. 2. Centre shows the outcrops of the pre-Tertiary inliers (stippled). The two collection areas are indicated. Top and bottom maps show the sampling sites. Unit 1 (lower diagram) is the Zion Hill granodiorite, the palaeomagnetic data for which will be reported elsewhere. All other units are Cretaceous dikes.

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

The island falls naturally into two provinces which are separated from each other by the Wagwater Belt. To the West, the Cretaceous succession consists largely of volcanics which were either sub-aerial, or deposited in shallow water and interbedded with limestones. Towards the end of the Cretaceous these rocks were deformed and intruded by granodiorite (Chubb & Burke 1963; Read 1966). There has been some discussion of associated metamorphism (Williams 1959; Burke 1964) but the event seems to be relatively minor and only of local significance. East of the Wagwater Belt the Cretaceous rocks appear to have been deposited in deeper water and consist of shale, sandstones, and volcanics. Chubb (1960) has suggested that they are geosynclinal. The closing Cretaceous upheaval was more marked here, and in some areas the rocks were intensely deformed and metamorphosed, reaching even the amphibolite facies. Granodiorites were intruded and serpentinites are associated with the event, reinforcing the suggestion of a geosynclinal area culminating in orogeny. Large scale volcanic activity ceased at the end of the Cretaceous.

The specimens collected for this study were mainly from a suite of dykes which intrude the volcanic sequence of the Central Inlier. The two major localities used (Fig. 2 upper) were Effort Bridge [grid reference 416 446] and the Rio Minho Gorge [between 436 448 and 444 442 on the Jamaica 1 : 50 000 sheet G]. In addition one site was sampled at New River (Fig. 2, lower), a tributary of the Rio Pedro. The cores were taken from a lamprophyre dike intruded into the Above Rocks granodiorite (Read 1966, p. 20 and 50), which is dated as uppermost Cretaceous (Chubb & Burke 1963).

The dikes of the central inlier intrude the formation known locally as the Nine Turns Conglomerate. This formation lies below the Rudist Limestones which are of Maastrictian age and so far no dikes have been found to cut these limestones. It is assumed therefore that the dikes are Pre-Maastrichtian, possibly Campanian or even Santonian in age. They are less deformed than the surrounding country rock which is often heavily fractured and veined, but they must have been involved in the late Cretaceous orogeny and subsequent earth movements. Whilst their strikes are variable they are all vertical which suggest that the disturbances have not resulted in any tilting. The possibility of regional rotation about a vertical axis cannot be resolved by field geological observations alone.

Experimental methods

Field

Up to six cores of average length 12 cm and diameter 2·5 cm were drilled from each of the 15 dikes using a portable petrol-powered drill. In nine instances geographic co-ordinates were used for orienting the cores to a precision of $\pm 3°$, but only between-core declination reference sightings could be made on the other six cases, necessitating use of the regional declination correction. When possible, comparison of the geographic sightings with this local declination correction provides confidence that a precision of no less than $\pm 4\frac{1}{2}°$ is obtained during the magnetic orientation.

Laboratory

Specimens of 2·2 cm length were sliced from each core, for palaeomagnetic measurement. Polished thin sections were made from slices adjacent to these specimens.

The natural remanent magnetism (NRM) directions and intensities were measured using a five cycles per second spinner magnetometer (Foster 1966). Unstable components were examined and minimized initially by applying alternating magnetic

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

fields (McElhinny 1966) of 100, 200, 300 and 400 oersteds peak field, to each specimen prior to remeasurement. Later, an attempt was made to improve such treatment by thermal demagnetizing methods, using a non-inductively wound oven with a $\pm 50\gamma$ ambient field, and nitrogen atmosphere, during controlled heating and cooling cycles up to 200 °C and 300 °C. All specimens were treated in identical ways. The mean direction of magnetism within each body was computed using a minimum scatter criterion. This computation method (Irving, Stott & Ward 1961) is preferred for several reasons (Watkins & Richardson 1968, 1970) over the more frequently used computation using only a single demagnetizing field treatment. Curie points were obtained for each specimen using a modified Chevallier balance and a rapid heating technique which inhibits thermomagnetic changes during the heating cycle.

The high temperature titanomagnetite oxidation index of four polished thin sections from each dike was determined optically using a Reichert Zetopan-pol reflection microscope, as described by Watkins & Haggerty (1967), and Watkins (1969). At the same time an estimate of the dominant grain-size fraction of the titanomagnetites present is made, and note is taken of the presence of maghemitization, which is low temperature alteration product, and ' granulation ' (or fine granular ferrirutile replacement of titanomagnetite) which Ade-Hall *et al.* (1968) believe is the result of reheating under considerable burial. The silicate petrography has been examined in each thin section using conventional transmitted light methods.

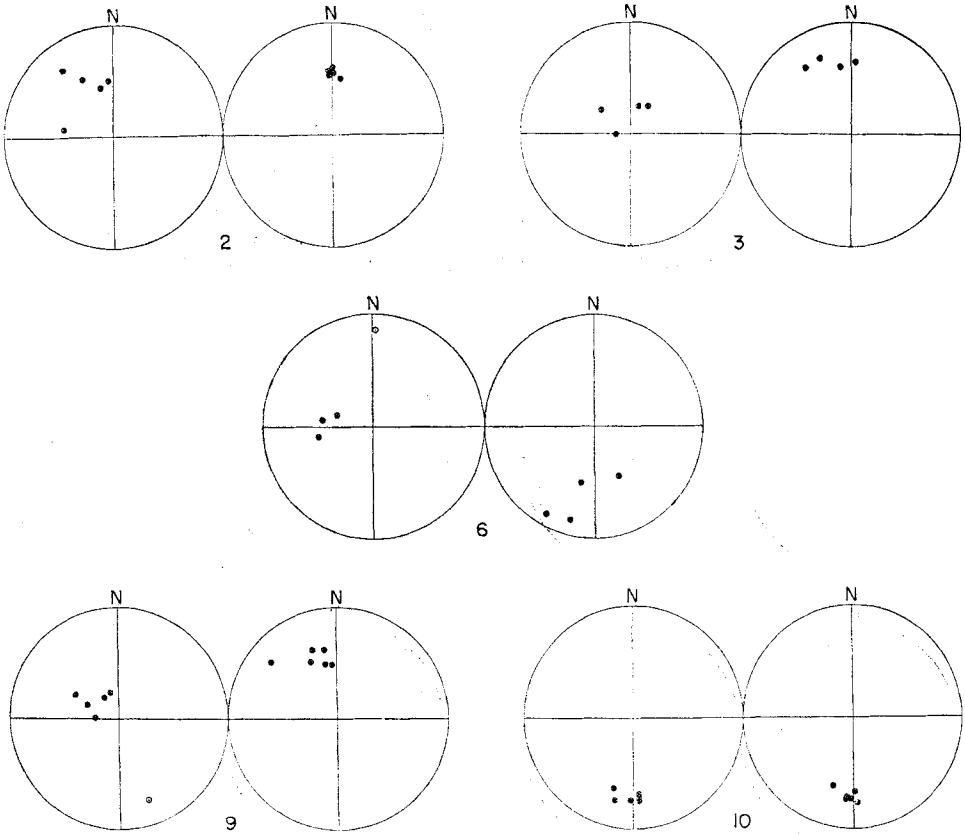


FIG. 3. Directions of remanent magnetism in dikes 2, 3, 6, 9 and 10 in each case before demagnetization (left) and following minimizing of unstable components by treatment in alternating magnetic fields (right). For discussion of demagnetization, see text. N = geographic north; $\otimes$ = lower hemisphere; $\odot$ = upper hemisphere. Equal area projections. See Figure 2 for dike locations.

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

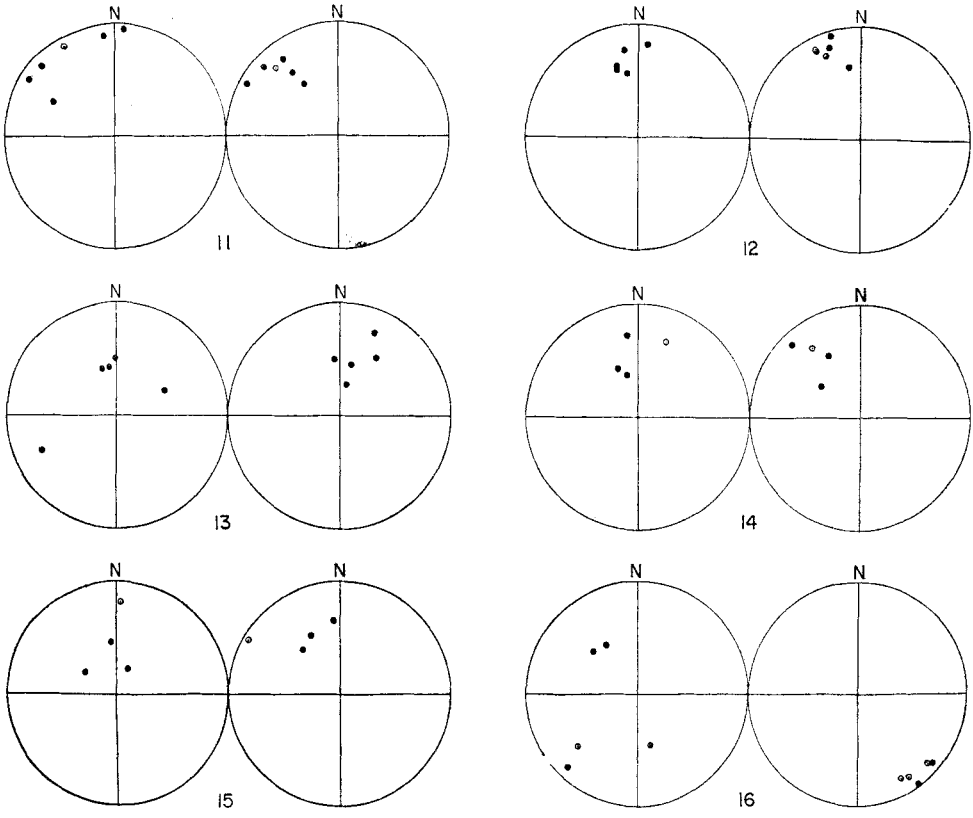


FIG. 4. Directions of remanent magnetism in dikes 11–16, in each case before demagnetization (left) and following minimizing of unstable components by treatment in alternating magnetic fields (right). See caption to Fig. 3 for further explanation.

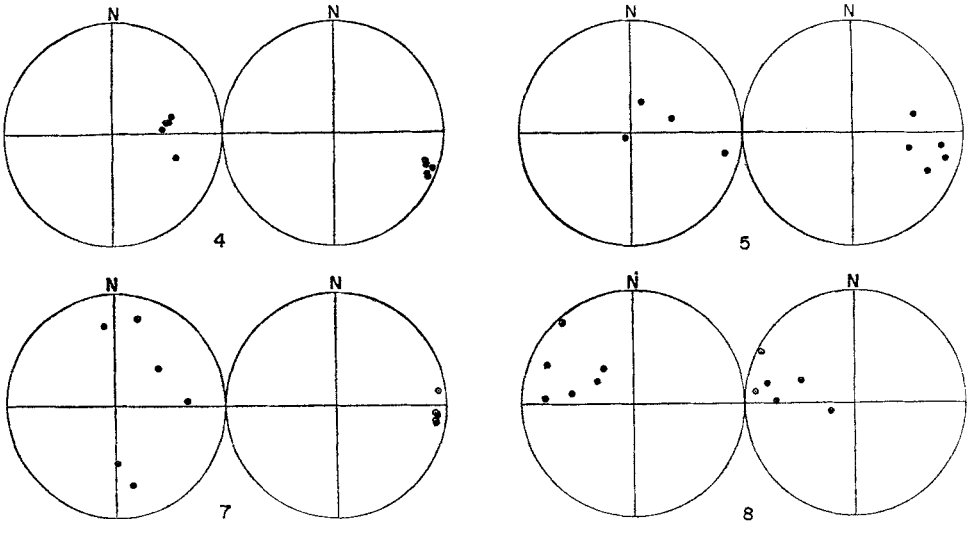


FIG. 5. Directions of remanent magnetism in units 4, 5, 7 and 8, in each case before demagnetization (left) and following minimizing of unstable components by treatment in alternating magnetic fields (right). See caption to Fig. 3 for further explanation.

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

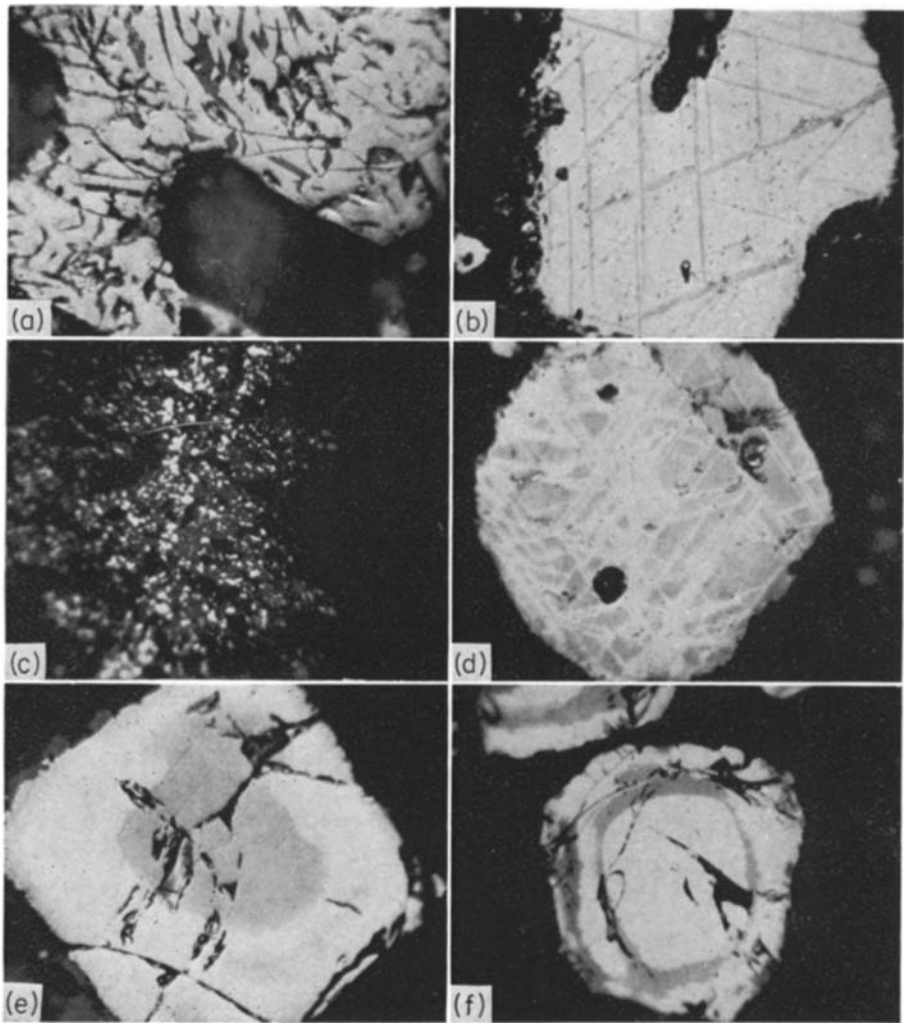


FIG. 8. Photomicrographs of titanomagnetites in the Jamaica dikes, taken using a Reichert Zetapan-pol reflection microscopes, and oil immersion. Field of view is 100×120 microns. Photomicrograph (a) shows dark ' granulation ' (Ade-Hall *et al.* 1968) development in titanomagnetite, probably after myrmekitic maghemite in core 5–2. (b) shows unusual development of titanomaghemite (white) after host titanomagnetite, with ilmenite lamellae remaining unoxidized and well defined in core 8–4. (c) shows development of very fine (approximately 1 micron) titanomagnetite blebs in an amphibole or pyroxene host in core 10–4. (d) shows fine titanomaghemite lamellae (white) development in titanomagnetite, with the centres of the lamallae still clearly unoxidized ilmenite, in core 12–4. (e) and (f) show a curious development of a concentric pattern, resulting from maghemitization (white) which develops on the fringes and centres of the grains so that a ring of residual non-maghemitized material is produced. This is clearly evident in (f), but only just perceptible in (e).

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

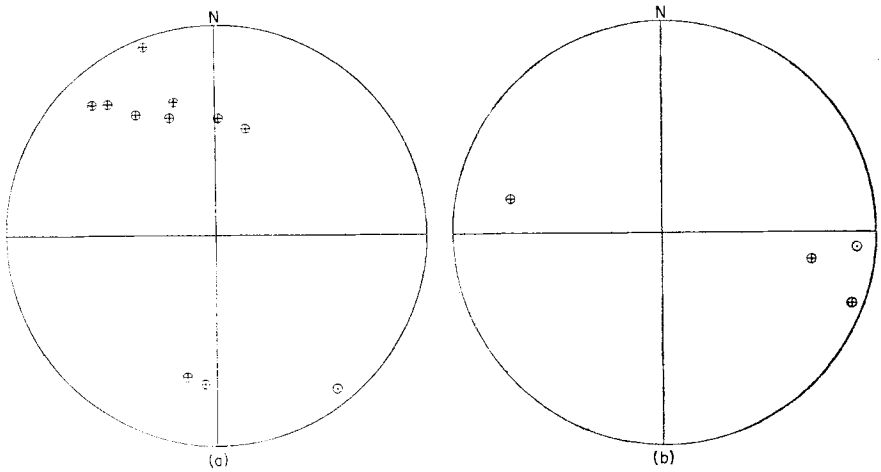


FIG. 6. Mean directions of remanent magnetism in all dikes, following minimizing of unstable components. The two groups, (a) and (b) are a natural subdivision of the data. See caption to Fig. 3 for further explanation.

Results

Figs 3–5 show the directions of magnetization in each specimen before and after minimizing unstable components. The corresponding mean directions of magnetization and other results of Fisher (1953) statistical analyses are shown in Fig. 6, and included in Table 1.

Fig. 7 shows four representative Curie point curves. All Curie point data are included in Table 2, which also shows the results of reflected and transmitted light examination of each polished thin section. In Fig. 8, six photomicrographs of the various titanomagnetites observed are shown.

Discussion

Magnetic stability and mineralogy

As with most dikes so far described in the literature, the titanomagnetites are dominated by the homogeneous lower oxidation state varieties, although some ilmenite lamellae are observed and in one case (14–1) haematized ilmenite lamellae are seen. Superimposed on these generally low oxidation states are three factors apparently related to observed magnetic differences.

The highest stability (Table 1, where $k > 100$) is observed in those dikes (numbers 2, 4, 7, and 10) which all feature very fine titanomagnetites, as small as 1 micron throughout (Fig. 8(c)), but no larger than 20 microns. Conversely, the lowest stabilities (Table 1, where $k < 10$) are observed in four dikes (numbers 6, 8, 14 and 15) featuring titanomagnetites as large as 200 microns. ' Granulation ' (Ade-Hall *et al.* 1968) is observed (Fig. 8(a)) on the larger grains as replacement after a myrmekitic titanomaghemite development. Ade-Hall *et al.* (1968) believe that deep burial of basaltic rock may result in ' granulation ' of titanomagnetites, and observe in one case that ilmenite occurrence and granulation are inversely related. Our detection of an inverse relationship of ' granulation ' with magnetic stability is probably fortuituous: the low magnetic stability depends more on the presence of large optically homogeneous titanomagnetites which are host to the ' granulation ' development. Maghemitization, although common in various forms in several dikes (Figs 8(b), (d), (e), (f)) and

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

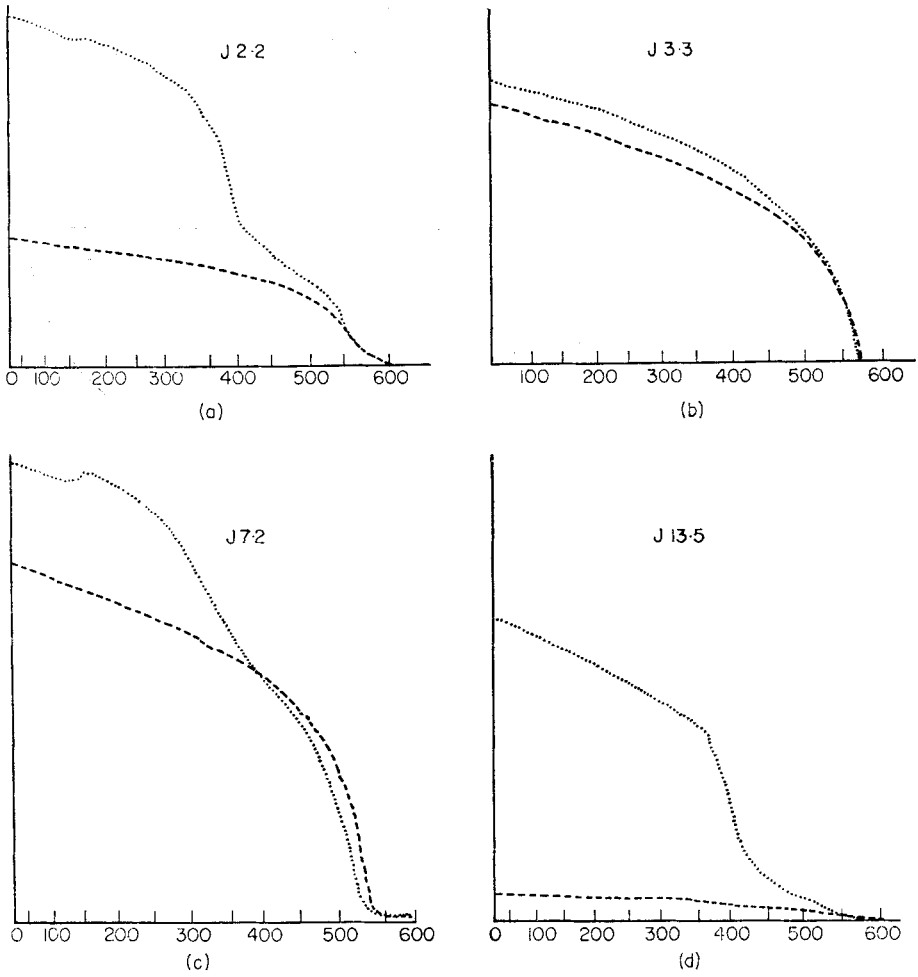


FIG. 7. Four examples of original Curie point curves. Specimen numbers as in Table 2. Dotted line is heating cycle; dashed line is cooling cycle. Absissae is temperature in degrees centigrade; ordinate is an arbitrary function of the saturation intensity of magnetization. Note the very large decrease of saturation intensity of magnetization in curves (a) and (d), on cooling. This behaviour can be diagnostic of the transformation of maghemite into haematite.

associated with low to moderate values of stability (dike numbers 5, 8, 13 and 15) appears to be subordinate to grain size in relationship to stability. That maghemite is dominant mineralogically in several dikes is supported by the Curie point data (Fig. 7(a), (d) and Table 2) where the heating treatment has converted maghemite into haematite, with a corresponding very large decrease of saturation intensity of magnetization, as shown on cooling. In fact, all 77 Curie point curves feature to some extent a loss of intensity of magnetization on cooling, very consistent with breakdown of maghemite to haematite in varying degrees.

An attempt to thermally remove unstable components in specimens not previously subjected to alternating magnetic field demagnetization was not successful. The specimens, sliced from positions in the cores adjacent the specimens examined earlier, were heated to 200 °C and cooled in a nitrogen atmosphere in a $\pm 50\gamma$ ambient magnetic field. Following measurement of resulting directions and intensities, the

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

Table 1

Palaeomagnetic Results

(a) Dikes providing high latitude virtual geomagnetic poles (Group A)

N°	N	$D°$	$I°$	K_0	K_{100}	K_{200}	K_{300}	K_{max}	ϕ'	θ'	$J_0\times10^4$	$\sigma J_0\times10^5$
2	5	001·5	+42·9	12·6	92·9	6·2	4·6	304·2	294·1	+82·9	2·77	16·4
3	4	343·8	+32·1	18·5	21·9	8·1	8·5	28·6	193·3	+74·6	3·21	8·4
6	4	190·6	+30·3	2·8	2·4	2·1	1·3	12·6	265·4	−54·2	2·67	5·4
9	6	338·4	+39·1	2·4	7·4	9·8	12·9	24·0	208·3	+69·8	1·58	3·4
10	6	183·7	+28·4	86·2	108·4	75·2	67·0	124·6	276·5	−56·7	5·65	24·7
11	6	316·6	+13·8	7·7	12·5	5·4	3·3	12·8	184·1	+46·3	3·04	5·9
12	6	399·1	+01·8	34·4	6·0	1·2	8·7	14·1	155·5	+63·3	3·14	6·0
13	5	016·7	+45·6	3·8	15·3	1·3	0·9	17·1	339·6	+72·1	3·57	16·1
14	4	321·6	+18·8	3·9	5·8	1·5	2·4	5·8	185·6	+51·9	2·23	1·3
15	4	326·3	+30·5	3·6	4·9	1·7	2·6	7·5	195·4	+57·8	2·16	7·0
16	5	141·4	−05·5	1·9	1·0	2·7	2·0	45·8	355·6	−49·2	1·06	5·5

(b) Dikes providing low latitude virtual geomagnetic poles (Group B)

N°	N	D	I	K_0	K_{100}	K_{200}	K_{300}	K_{max}	ϕ'	θ'	$J_0\times10^4$	$\sigma J_0\times10^5$
4	5	110·9	+00·7	48·8	63·6	3·0	2·1	117·4	5·9	−19·7	3·89	4·3
5	5	100·4	+27·1	2·5	2·3	2·6	3·6	17·2	356·1	−05·2	1·79	4·1
7	6	094·5	−05·1	1·7	94·3	33·4	5·7	155·7	14·0	−5·1	5·16	9·5
8	6	282·1	+25·9	8·1	5·8	1·3	1·6	6·6	202·6	+15·4	4·05	17·5

(c) Combined data for each group

Group	B	N	D	I	K	ϕ'	θ'	α_{95}	dp	dm
A	11	55	341·9	+17·1	6·3	167·7	70·1	19·8	10·0	20·5
B	4	22	102·0	+ 0·9	12·8	9·1	−11·6	26·7	13·4	26·7

In (a) and (b): No. = site number (Fig. 2); N = number of cores per site; $D°$ and $I°$ = declination and inclination of remanent magnetism, following demagnetization (D is east of north, I is with respect to the horizontal, + = downward, − = upward); $K = N-1/N-R$, where R = resultant vector applying unit vector per core; suffices 100–300 to K are magnitude of alternating magnetic field treatment; K_{max} = K for minimum dispersion computation (Irving *et al.* 1961); ϕ' and θ' are longitude and latitude of virtual geomagnetic poles, ϕ' is East longitude, θ' is + for Northern Hemisphere, − for Southern Hemisphere; J_0 = intensity of natural remanent magnetism in emu g^{-1}, σJ_0 = standard deviation of J_0. In (c): as in (a) and (b), except for: B = number of dikes; N = total number of cores; $K = B-1/B-R$, where R = resultant vector applying unit vector per dike; α_{95} = semi vertical angle of confidence core; dp and dm = angular length semi-axis of 95 per cent confidence oval, along and normal to the great circle joining the sample site and virtual geomagnetic pole.

Table 2

Opaque mineralogical data, Curie points, silicate petrology

Dike No.	Core No.	Ox. index	Grain size(s) (microns)	G	S	T_c	T_c type	Petrological type	Comments
2	1	IM	10			500	1b	Lamprophyre	
	2	IM	10		×	400, 575	3b	,,	
	3	IM	10		×	150, 540	3b	,,	
	4	IIM	10			400, 575	3b	,,	
3	1	I	<5, 50		×	575	1b	Trachyandesite	
	2	I	<5, 50			575	1	,,	
	3	?	30			565	1b	,,	
	4	III	<5, 20			575	1b	,,	

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

Table 2 (*continued*)

4	1	II	20			580	1	Porphyritic trachybasalt
	2	I	< 5, 20		×	575	1	,,
	3	I	< 5, 20			540	1b	,,
	4	I	40			575	1	,,
5	1	IM	10, 150	×		510	1b	' Altered ' basalt
	2	?	10, 150	×		540	1b	,,
	3	?	10, 150	×		560	1	,,
	4	IM	20, 100	×		585	1b	,,
6	1	II	30	×		490	1	Basalt
	2	I	20	×		555	1b	,,
	3	I	< 5, 30, 100	×		550	1b	,,
	4	I	30	×	×	535	1b	,,
7	1	I	20			550	1	Hornblende andesite
	2	IM	20			150, 540	IIIb	,,
	3	II	20	×	×	570	1	,,
	4	II	20	×		555	1	,,
8	1	IM	20, 100		×	500	1	Dolerite
	2	?	20, 100			540	1b	,,
	3	IIM	200	×		475	1b	,,
	4	IM	150			535	1b	,,
9	1	I	60			555	1	Hornblende andesite
	2	I	100			560	1b	,,
	3	IIM	10, 100			575	Ib	,,
	4	I	< 5, 100			575	1b	,,
10	1	?	< 5	×		560	1b	,,
	2	?	< 5			540	1b	,,
	3	?	< 5			570	1	,,
	4	?	< 5, 50			560	1	,,
11	1	V	15			575	1b	Trachyandesite
	2	V	15, 80			585	1b	,,
	3	I	80		×	535	1b	,,
	4	IM	10			375, 575	3b	,,
12	1	?	20			545	1b	Porphyritic andesite
	2	?	100			525	1b	,,
	3	IM	150			575	1b	,,
	4	IM	200			525	1b	,,
13	1	IM	5, 100			425, 575	3b	Hornblende andesite
	2	IM	10, 100		×	550	1b	,,
	3	IM	15, 150	×		560	1b	,,
	4	IM	20, 100	×		540	1b	,,
14	1	V	20			525	1b	Hornblende andesite
	2	I	10, 60	×		510	1b	,,
	3	IM	10, 150			535	1b	,,
	4	?	5			570	1b	,,
15	1	IM	100	×		555	1	Hornblende andesite
	2	I	80		×	525	1b	,,
	3	I	60			500	1b	,,
	4	IM	20		×	540	1b	,,
16	1	IM	15, 150			525	1	Hornblende andesite
	2	I	15, 150			525	1b	,,
	3	I	20, 150			530	1b	,,
	4	I	40, 100			525	1b	,,

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

Table 2 (*continued*)

Oxidation Index: based on estimate following scan of 100 or more titanomagnetite grains. Index I = optically homogeneous titanomagnetite; II = rare ilmenite lamellae in titanomagnetite; III = common ilmenite lamellae in titanomagnetite; IV = haematized lamellae after ilmenite in titanomagnetite; *V* = pseudobrookite-hematite intergrowths. Suffix *M* = substantial fraction of maghemitized titanomagnetites, ? = indeterminable index, because of very mixed indexes or too fine for optical identification. For additional diagnostic criteria, see Watkins & Haggerty (1967) or Watkins (1969). Grain size estimated from optically scanning of dominant opaque fraction. This is sometimes bi or trimodal. *G* = titanomagnetite granulation present. [This is a ferrirutile granulation which is thought to result from secondary heating as a result of burial (Ade-Hall *et al.* 1968)] *S* = fine sulphides present (*X* in *G* or *S* column indicates presence; blank indicates absence of *G* or *S*) T_c = Curie point(s), in degrees centigrade. T_c type is either 1 (simple single knee curve, consistent with dominance of single constituent); 2 (curve without knee; consistent with variable constituents, with a single dominant component); or 3 (curve with two knees, consistent with either a high and a low Curie point subsequently modified during heating). Suffix to T_c type is either a (cooling curve above heating curve), or b (cooling curve below heating curve). No suffix to T_c type indicates heating and cooling curves virtually identical. Petrological type results from transmitted thin section analysis. Note: doubt exists concerning the exact identification of index *V* grains of dikes 11 and 14.

operation was repeated at 300 °C. In both cases, the directions were virtually random, except in dikes 2, 4 and 10, and the intensities of magnetization were decreased by two orders of magnitude compared to the untreated state. This is consistent in part with the transformation of maghemite to haematite.

It is not suggested that the alternating magnetic field cleaning, as applied to all specimens, has resulted in perfect removal of the unstable components which are clearly present in many specimens (Figs 3–5), but the data represent our best objective experimental approach to arriving at the most acceptable result.

In Fig. 6, the mean directions of magnetization following demagnetization are presented in two groups corresponding to the natural subdivision of the data. The larger group (Fig. 6(a)) is non-Fisherian in its distribution, tending to a 'streaking' character. We clearly cannot selectively reject any part of the data, and profitable further demagnetization is not possible in our specimens. In any case, there is no particular reason to be certain that the particular Cretaceous geomagnetic field involved did not have a behaviour generally corresponding to the directions measured.

Directions of magnetization

The coercivity spectra within and between each dike is variable, and can be related to mineralogical differences as in the above discussion of stability variation. There are no systematic magnetic or thermomagnetic differences, however, between the two clearly different groups of data given in Fig. 6. Polarity changes are found in both data sets. The data in both groups are from well-separated sites. Fig. 6(a) includes data from the upper and lower Minho River sites. In Fig. 6(b) the easterly declination group is from the Effort Bridge section, and the westerly declination is from the Upper Rio Minho site. We conclude that the separation of the data into two groups reflects two real stable palaeomagnetic directions and not fortuitous sampling of materials with different stabilities, or the result of systematic experimental inadequacies.

Prior to stating with any confidence that Cretaceous geomagnetic field is represented by the two data groups, we must examine the possibility that a thermal overprint and tectonic rotation, or large age differences within the collection, can have caused the separation of a single data group into the two groups detected. In Fig. 9, a model is presented which would explain the separation of the data, if the small inclination differences between each group are neglected. Reheating of the larger data group following very late Cretaceous rotation, with the smaller data group retaining the original Cretaceous direction which is now rotated into an anomalous direction, is illustrated. This model, which is capable of integration into a previous

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

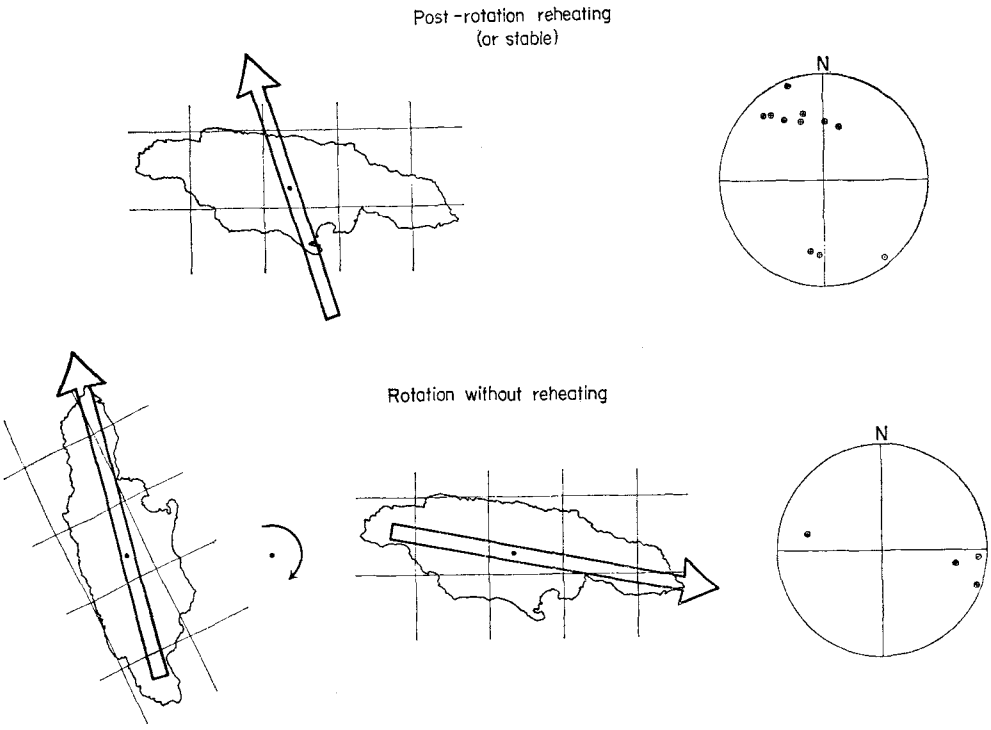


FIG. 9. A model hypothesizing the observed separate grouping of remanent magnetism in the Jamaican dikes (Fig. 6) in terms of post-cooling modification. The large open arrow represents the paleomagnetic declination direction. The larger data group (top right) would need to have been thermally-overprinted in a Cretaceous geomagnetic field direction, following large rotation of the island about a vertical axis. The smaller data group (bottom right) would need to have preserved the original Cretaceous geomagnetic field, and have been rotated into their present directions (without any thermal overprint) by large-scale rotation of Jamaica about a vertical axis. See text for discussion of the merits of this model.

suggestion [the 150° clockwise rotation of Cary 1958, Fig. 1(a)] is not considered as feasible since no petrological, thermomagnetic, or field data shows that the larger data group have been reheated. A second possibility is that the smaller data group are substantially older than the unrotated larger data set, and experienced rotation prior to the emplacement of the latter. Field relationships do not readily support this possibility. As will be shown below, the smaller group result in equatorial VGP. A rotation to create equatorial VGP would be an unlikely but not impossible coincidence. The two data groups are interpreted by us to represent different Cretacous geomagnetic field directions: this is supported by some field and petrological observations. The smaller group of Fig. 6(b) includes two dikes at Effort Bridge which are porphyritic, and are therefore different in age from the other bodies in the same section, since these are generally non-porphyritic. Emplacement characteristics of dikes 4 and 5 also differ to the other dikes at Effort Bridge: they actually cut each other and are irregular and of anomalous trend.

Interpretation

The directions of magnetization shown in Figs 6(a) and (b) correspond to the VGP shown in Figs 10 and 11 respectively. It is emphasized that the resulting mean

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

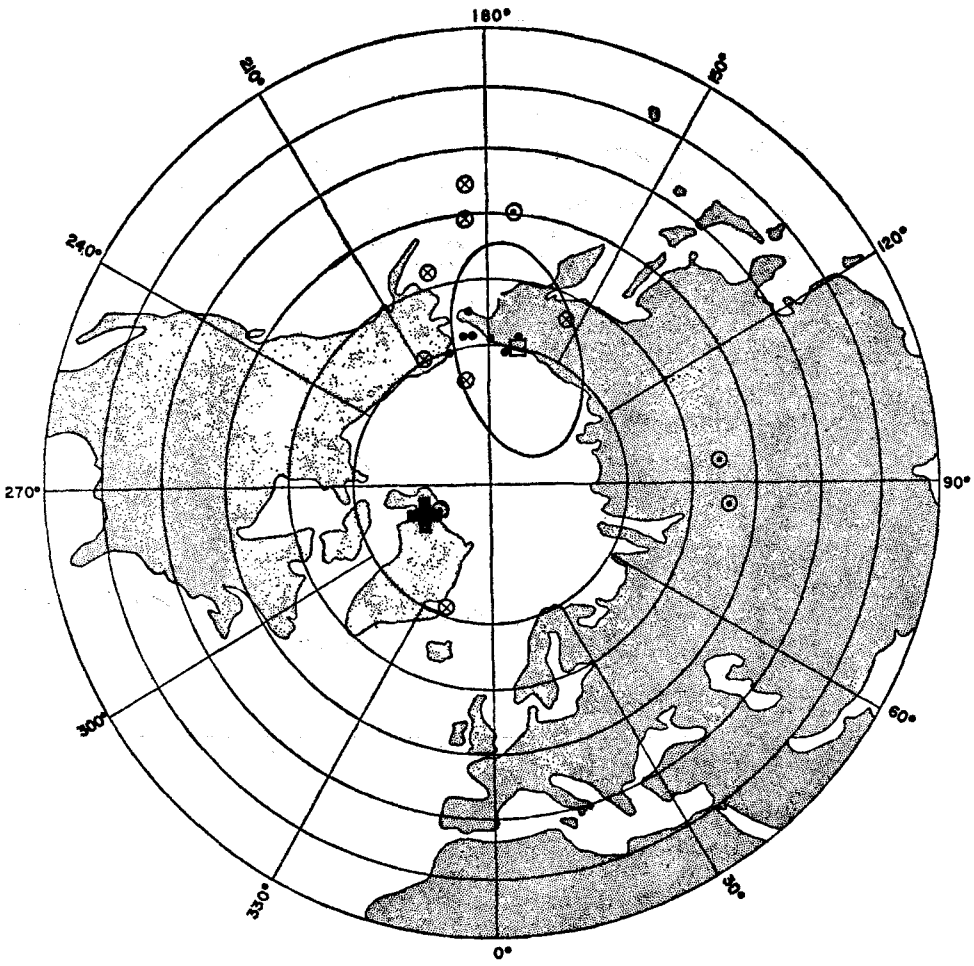


FIG. 10. Virtual geomagnetic poles for each dike of the larger data group (Fig. 6). ⊙ = reversed polarity, transposed to the Northern Hemisphere; ⊗ = normal polarity. + = present geomagnetic pole. ⊡ = mean VGP position, (the 95 per cent confidence oval is included) ● = previously reported VGP for seven studies of the North American Cretaceous (Larochelle 1968).

VGP and oval of confidence in Fig. 10 represents the taking of some statistical liberties, in that the required Fisherian distribution (analogous to Gaussian distribution, but on a sphere) is not present, but no other presentation convention is meaningful in the context of the desired comparison with other Cretaceous palaeomagnetic data from North America. The oval of confidence encloses the best estimate of the stable North American Cretaceous pole position, which is in the vicinity of the northwest side of the Bering Straits (Irving 1964; Larochelle 1968), as shown in Fig. 10. It is considered highly unlikely that this similarity is fortuitous, being caused by field and laboratory experimental imperfections. The most reasonable (if not the only) interpretation of our larger data group is, therefore, that no large rotation of Jamaica about a vertical axis has occurred since the Cretaceous. Even if the smaller data group (Fig. 6(*b*) is shown to be substantially older than the larger data group (Fig. 6(a)) allowing an explanation for the separation of the two data groups along tectonic lines similar to Carey's model (Fig. 1(a), Fig. 9, lower), this conclusion is unaffected: no significant *post*-Cretaceous rotation is indicated. The possibility of

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

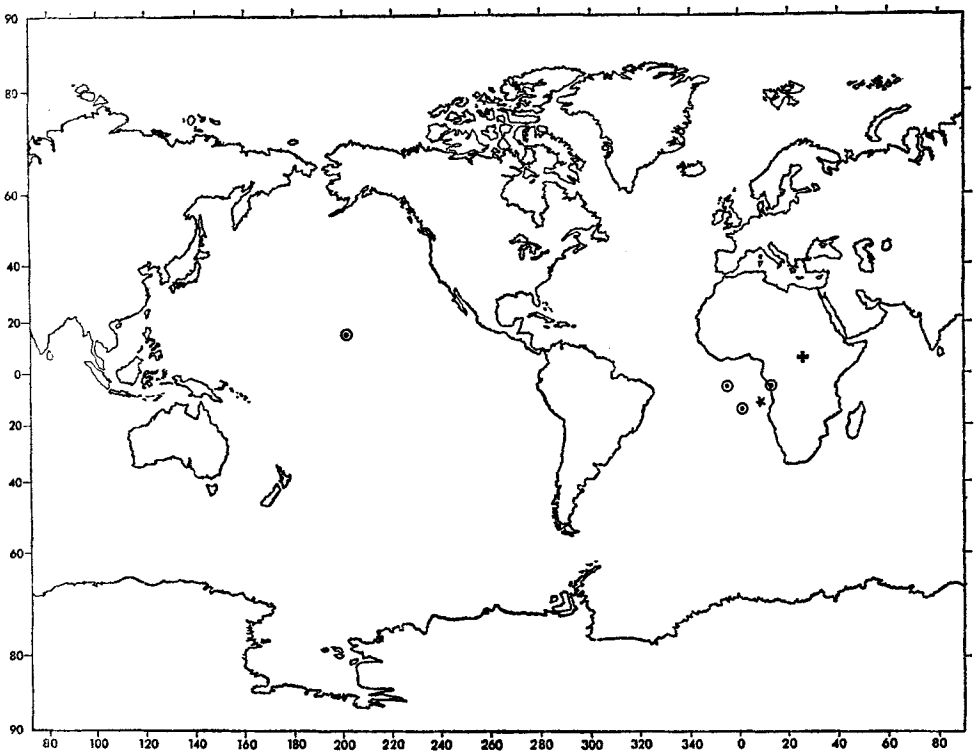


FIG. 11. Virtual geomagnetic poles (⊙) for the four dikes which constituting the smaller data group (Fig. 6). ★ = mean VGP (after transposing the Pacific VGP). + = VGP reported by MacDonald & Opdyke (1970) for the Lower Cretaceous of Columbia, South America.

an east–west translation of up to 1000 km is, however, not excluded. This simply cannot be detected by data with the confidence parameter which we define.

In Fig. 11, equatorial VGP are defined, resulting from dikes 4, 5, 7 and 8. Equatorial VGP have been previously detected in several other areas in Tertiary basalts: Alaska (Cox 1966); Hawaii (Doell 1968); Iceland (Dagley *et al.* 1967); and Oregon (Watkins 1969).

Cox (1966) explains his data, from lavas of Brunhes (0 to 0·69 My) age as due to non-dipole activity, and Doell's (1968) results, also from young lavas, appear to represent main dipole movement. The data from Oregon (Watkins 1969) are part of a Miocene geomagnetic polarity transition, and many of the data from Iceland (Dagley *et al.* 1967) appear to be related to polarity changes. It is therefore not surprising that equatorial VGP have been detected in Cretaceous rocks, although none of the other detections of equatorial VGP report a 'reversal' (or 180° degree separation of direction) within the equatorial VGP group. This suggests that either a clearly non-axial dipole configuration persisted stably for a period sufficient for polarity change to occur, or different non-dipole dominated directions occurred at different times, and by coincidence yield VGP with a 180° separation. Multipole mechanisms could also be invoked. Additional studies of Cretaceous rocks could resolve these ambiguities.

It is interesting to compare the mean VGP obtained from our 'anomalous' Jamaican dikes ($\phi' = 9{\cdot}1°$ E, $\theta' = 11{\cdot}5°$ S) with the unusual mean VGP recently reported by MacDonald & Opdyke (1970) for the Cretaceous of North Colombia. Their VGP is near $\phi' = 26°$ E, $\theta' = 6°$ N. This similarity with our result (Fig. 11)

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

is remarkable, particularly when the highly unusual nature of the VGP positions is considered. In their case, however, the age reported for the rocks studied is Lower Cretaceous ($t = \sim 120$ My), and therefore it is unlikely that the same geomagnetic field behaviour is reported in both studies. For this to be the case, either the four Jamaica dikes sampled would need to be older than inferred from the field relationships, and/or the Colombia study would need to have been made on rocks younger than Lower Cretaceous.

Conclusions

Our study of 15 Cretaceous dikes from the island of Jamaica has provided the following results:

1. The titanomagnetites are of a generally lower oxidation, optically homogeneous, type. Maghemitization is detected optically in many specimens, and is confirmed thermomagnetically. The Curie point curves *all* feature loss of saturation magnetization on heating, consistent with titanomaghemite being present in all specimens to some extent. The observed stability variations are due dominantly to grain size variation. Maghemitization may also have reduced stability, and 'granulation' (secondary development of ferri-rutile granulation in optically homogeneous titanomagnetite) appears in low stability specimens. None of the palaeomagnetic directions and intensities in the dikes are interpreted to represent functions of reheating or tilting, or to be drastically different from the ambient geomagnetic field during the initial cooling.

2. Eleven of the dikes have a high latitude VGP. The distribution of the VGP is somewhat streaked suggesting either imperfect removal of unstable components or sampling of a corresponding Cretaceous geomagnetic field behaviour. When averaged the directions cannot be distinguished from a Cretaceous geomagnetic field direction resulting from the Cretaceous geomagnetic pole of North America. Our favoured interpretation for the data set is that Jamaica has not rotated significantly about a vertical axis since the Cretaceous. While we acknowledge that further collecting may yield a Cretaceous pole with a smaller confidence interval, we do not expect that this conclusion can be drastically modified.

3. The remaining four dikes provide low latitude VGP. They are generally magnetically stable and closely grouped ($k = 12{\cdot}8$), despite the fact that age differences exist between the dikes, as shown by the fact that one dike is reversely magnetized relative to the other three. Tectonism and/or reheating is not favoured as explaining the difference in direction between the two data groups. Equatorial VGP have been detected in the Tertiary of Alaska, Hawaii, Iceland and Oregon. These Jamaica data are interpreted to represent unusual geomagnetic field behaviour, which results in directions very similar to recent Lower Cretaceous data from Colombia. This conclusion could be modified if it is ever shown that the four dikes are substantially older than the larger data group.

4. According to our data, a normal polarity geomagnetic field dominates (but not completely so), during the Upper Cretaceous as predicted by Helsley & Steiner (1969). If the compilation of Helsley & Steiner (1969, Fig. 1) is complete, our reversely magnetized dikes could be either Maestrichtian or Santonian in age, rather than Campanian as originally suspected. Alternatively, we may have detected the first Campanian reversed polarities.

Acknowledgments

The field work was made in conjunction with Dr A. Richardson, Department of Geophysics, Imperial College, London. Laboratory measurements were made

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

using apparatus funded in part by the National Science Foundation (grant numbers GA 1123, GA 1620, and GA 13132).

N. D. Watkins:
Graduate School of Oceanography,
Narragansett Marine Laboratory,
University of Rhode Island,
Kingston, Rhode Island 02881

F. W. Cambray:
Department of Geology,
Wayne State University,
Detroit, Michigan 48202.

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026

References

Ade-Hall, J., Khan, M. A., Dagley, P. & Wilson, R. L., 1968. A detailed opaque petrological and magnetic investigation of a single Tertiary lava flow from Skye, Scotland-II spatial variations of magnetic properties and selected relationships between magnetic and opaque petrological properties, *Geophys. J. R. astr. Soc.*, **16,** 389.

Burke, K., 1964. Letter to the editor, *Geonotes* (*Jamaica Geol. Soc.*), **6,** 46.

Carey, S. W., 1958. The tectonic approach to continental drift. *Continental Drift, A Symposium,* p. 177. University of Hobert.

Chubb, L. J., 1960. Geosynclines and epicontinental seas, *J. geol. Soc. Jamaica,* **3,** pp. 2 and 53.

Chubb, L. J., 1962. Below the Cretaceous, what? *J. geol. Soc. Jamaica,* **5,** 2.

Chubb, L. J. & Burke, K., 1963. Age of the Jamaican Granodiorite, *Geol. Mag.*, **100,** 524.

Cox, A. V., 1966. Geomagnetic secular variation in Alaska. *Trans. Am. Geophys. Un.*, **47,** 78.

Dagley, P., Wilson, R. L., Ade-Hall, J., Walker, G. P. L., Haggerty, S. E., Watkins, N. D., Smith, P. J., Edwards, J. & Grasty, R., 1967. Geomagnetic polarity zones for Icelandic lavas, *Nature*, **216,** 25.

Doell, R. R., 1968. Paleomagnetic studies of lavas of the islands of Kauai and Oahu, *Trans. Am. Geophys. Un.*, **49,** 127.

Fisher, R., 1953. Dispersion on a sphere, *Proc. R. Soc. London*, **A217,** 295.

Foster, J. H., 1966. A paleomagnetic spinner magnetometer using a fluxgate gradiometer, *Earth planet Sci. Lett.*, **3,** 48.

Gough, I. & Heirtzler, J. R., 1969. Magnetic anomalies and tectonics of the Cayman Trough, *Geophys. J. R. astr. Soc.*, **18,** 33–44.

Heirtzler, J. R., Dickson, G. O., Herron, E. M., Pitman, W. C. & LePichon, X., 1968. Marine magnetic anomalies, geomagnetic field reversals, and motions of the ocean floor and continents, *J. geophys. Res.*, **73,** 2119.

Helsey, C. E. & Steiner, M. B., 1969. Evidence for long intervals of normal polarity during the Cretaceous period, *Earth planet Sci. Lett.*, **5,** 325.

Hess, H. & Maxwell, J. C., 1953. Caribbean research project, *Bull. geol. Soc. Am.*, **64,** 1–6.

Irving, E., 1964. *Paleomagnetism and its application to geological and geophysical problems*, 399 pp., John Wiley, New York.

Irving, E., Stott, P. M. & Ward, M. A., 1961. Demagnetization of igneous rocks by alternating magnetic fields, *Phil. Mag.*, **6,** 549.

Larochelle, A., 1968. Paleomagnetism of the Monteregian Hills: new results, *J. geophys. Res.*, **73,** 3239.

MacDonald, W. D., 1968. Continental drift and the Caribbean (Deplazamiento continental y el Caribe), *Bol. Inf. Geol. Assoc. Venezuela*, **11,** 17.

MacDonald, W. D. & Opdyke, N. D., 1970. A Cretaceous paleomagnetic pole position for northernmost South America, *Trans. Am. Geophys. Un.*, **51,** 271.

McElhinny, M. W., 1966. An improved method for demagnetizing rocks in alternating magnetic fields, *Geophys. J. R. astr. Soc.*, **10,** 369.

Molnar, P. & Sykes, L. R., 1969. Tectonics of the Caribbean and middle America regions from focal mechanisms and seismicity, *Geol. Soc. Am. Bull.*, **9,** 1639.

North, F. K., 1965. The curvature of the Antilles, *Geol. en Mijn.*, **44,** 73.

Read, A. H., 1966. Geology of the Bog Walk quadrangle, Jamaica, *Bull. Jamaica geol. Survey (Kingston)*, **6,** 54.

Rod, E., 1967. Paleotectonic reconstruction of the Antilliean-Caribbean area for the close of the Carboniferous, *Bol. Inf. Geol. Assoc. Venezuela*, **10,** 196.

Watkins, N. D., 1969. Non-dipole behaviour during an Upper Miocene geomagnetic polarity transition in Oregon, *Geophys. J. R. astr. Soc.*, **17,** 121.

Watkins, N. D. & Haggerty, S. E., 1967. Primary oxidation variation and petrogenesis in a single lava, *Contrib. mineral. Petrol.*, **15,** 251.

Watkins, N. D. & Richardson, A., 1968. Palaeomagnetism of the Lisbon Volcanics, *Geophys. J. R. astr. Soc.*, **15,** 287.

Watkins, N. D. & Richardson, A., 1970. Reply to comments by K. M. Storevedt on ' Palaeomagnetism of the Lisbon Volcanics ' (by Watkins, N. D., & Richardson, A.), *Geophys. J. R. astr. Soc.*, (in press).

Williams, J. B., 1959. Field meeting in the Central Inlier of Jamaica, *Proc. Geol. Assoc.*, **70,** 254.

Zans, V. A., Chubb, L. J., Versey, H. R., Williams, J. B., Robinson, E. & Cooke, D. L., 1962. Synopsis of the geology of Jamaica, *Bull. Jamaica geol. Surv.*, **4,** 70 pp.

Downloaded from https://academic.oup.com/gji/article/22/2/163/550953 by Curtin University Library user on 06 May 2026